

ARC 通用人工智能挑战赛银牌算法概览

1. 本次比赛旨在解决小样本条件下的逻辑推理问题，属于人工通用智能（AGI）领域，旨在测试人工智能系统学习新技能的能力。目前，AI 在这一方面的能力仍远远低于人类。比赛结果将通过准确率进行评估。
2. 由于比赛没有训练数据，我们无法使用传统机器学习方法来训练模型。因此，我们采用了一系列启发式 DSL 算法，使模型具备处理未见场景的能力。具体步骤包括：
 - 任务生成：创建 Task 对象并完成预处理；
 - 变换：变换原始任务，可能变换包括旋转、填充、翻转、移动、上色等；
 - 候选者生成：生成多个虚拟候选者，并在循环中优化，尝试各种操作以提高得分；
 - 得分计算：通过比较像素的正确性来计算得分；
 - 最终候选者：还原操作以获得最终候选者，并可能生成新任务以增加样本量并比较候选者。
3. 为了提高模型的稳健性，我们通过比赛提供的验证集和交叉验证方法，反复调整启发式算法中的参数，充分测试模型的泛化能力，以避免过拟合。这一系列方法表现出了卓越的性能，适应了本次比赛的数据特点。
4. 在比赛中，我们充分利用了时间，合理控制了比赛的节奏；积极讨论并制定了合理的比赛模型迭代任务，并组织了头脑风暴来推进比赛进展。同时，我们着手进行了模型的开发工作，以尽可能提高预测准确性。最终，我们取得了 top x% 的排名。

Silver Medal Algorithm Overview for ARC Prize 2024

1. This competition aims to address logical reasoning problems under small sample conditions, which falls within the field of Artificial General Intelligence (AGI). Its purpose is to test the ability of AI systems to learn new skills, a capability in which current AI still lags far behind humans. The results of the competition will be evaluated based on accuracy.

2. Since there is no training data provided for the competition, we cannot use traditional machine learning methods to train our models. Therefore, we employ a series of heuristic algorithms to enable the model to handle unseen scenarios. The specific steps include:

- Task Generation: Creating a Task object and completing preprocessing;
- Transformation: Transform the original task, which may include rotation, filling, flipping, moving, coloring, etc.;
- Candidate Generation: Generating multi dummy candidates and optimizing them in a loop, trying various operations to improve scores;
- Score Calculation: Calculating scores by comparing pixel correctness;
- Final Candidates: Reverting operations to obtain the final candidates, and possibly generating new tasks to increase sample size and compare candidates.

3. To enhance the robustness of the model, we repeatedly adjust the parameters in the heuristic algorithms through the validation set and cross-validation methods provided by the competition, thoroughly testing the model's generalization ability to avoid overfitting. This series of methods has demonstrated outstanding performance and is well-suited to the data characteristics of this competition.

4. Throughout the competition, we made full use of time and managed the competition pace effectively; we actively discussed and developed reasonable model iteration tasks and organized brainstorming sessions to advance the competition's progress. At the same time, we focused on model development to maximize prediction accuracy. In the end, we achieved a ranking in the top x%.